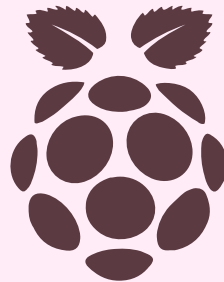


PeopleTec Pi Day Competition

Raspberry Pi Project

Summer Phillips



Hardware

Raspberry Pi 3 Model B



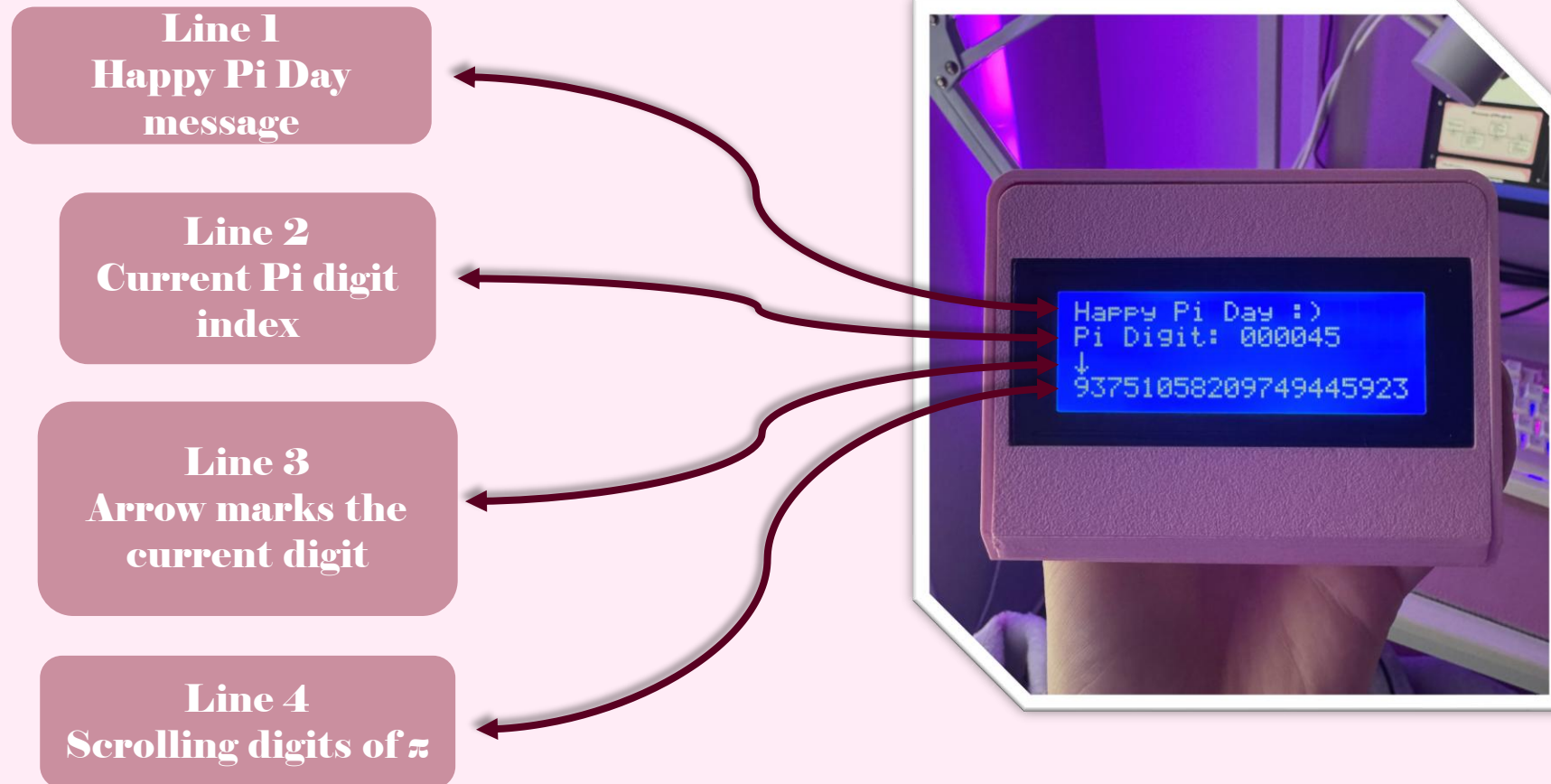
**20x4 LCD Display
with I2C backpack**



3D printed case

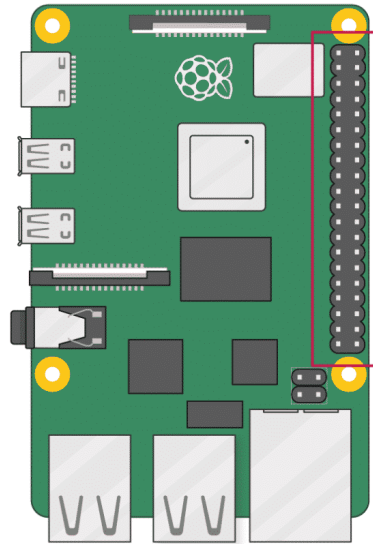


Demonstration



The display refreshes every 100 digits and briefly shows the “Happy Pi Day” message.

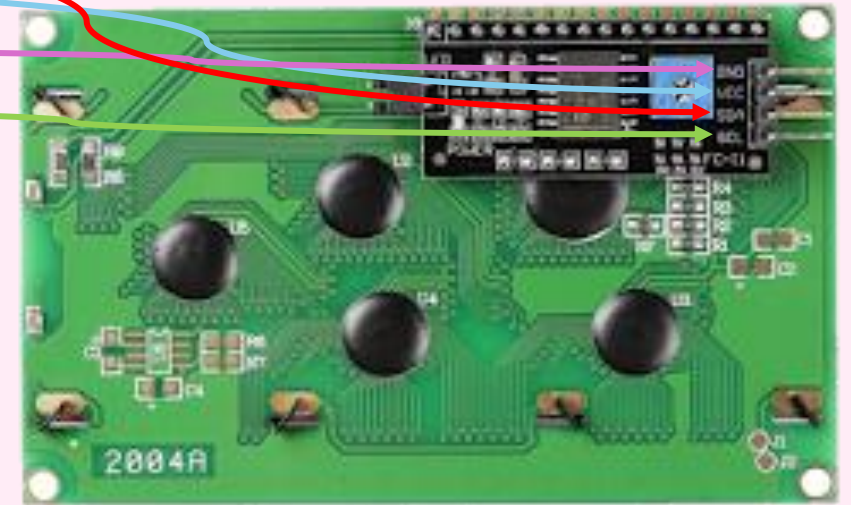
Process of Project - Hardware



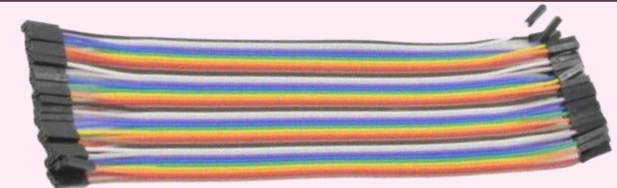
3V3 power	1	2	5V power
GPIO 2 (SDA)	3	4	5V power
GPIO 3 (SCL)	5	6	Ground
GPIO 4 (GPCLK0)	7	8	GPIO 14 (TXD)
Ground	9	10	GPIO 15 (RXD)
GPIO 17	11	12	GPIO 18 (PCM_CLK)
GPIO 27	13	14	Ground
GPIO 22	15	16	GPIO 23
3V3 power	17	18	GPIO 24
GPIO 10 (MOSI)	19	20	Ground
GPIO 9 (MISO)	21	22	GPIO 25
GPIO 11 (SCLK)	23	24	GPIO 8 (CE0)
Ground	25	26	GPIO 7 (CE1)
GPIO 0 (ID_SD)	27	28	GPIO 1 (ID_SC)
GPIO 5	29	30	Ground
GPIO 6	31	32	GPIO 12 (PWM0)
GPIO 13 (PWM1)	33	34	Ground
GPIO 19 (PCM_FS)	35	36	GPIO 16
GPIO 26	37	38	GPIO 20 (PCM_DIN)
Ground	39	40	GPIO 21 (PCM_DOUT)

Raspberry Pi pin Layout

LCD Screen



Dupont Jumper Cables (Female to Female)



Process of Project

Hardware
Wired LCD display
to Raspberry Pi

Step 2

Programming
Used ChatGPT
to help write
Python script to
control LCD

Step 4

Step 1

Communication
Enabled I2C protocol
Installed I2C
diagnostic tools
Verified LCD
detection

Step 3

Automation
Created a systemd
service
Program runs
automatically when
device powers on

Challenges

1.

Problem:
Flickering Screen

Cause:
Issues keeping the jumper
wires firmly connected

Solution:
Hot glue the wires to the
raspberry pi base

2.

Problem:
Configuring the I2C

Cause:
The I2C interface was not
detected

Solution:
Enable the I2C protocol in
the settings

The End



This presentation ends here... but Pi keeps going.